

TOPIC NAME: Regression Tree

DAY: _____

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Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

$$\begin{aligned}
 &+ (252.12 - 001) + (252.12 - 001) \\
 &+ (252.12 - 05) + (252.12 - 001) \\
 &\quad \text{male} \quad \text{female} \\
 &\quad 0, 2, 100, 100, 0 \quad 3, 100, 20, 10
 \end{aligned}$$

$$\begin{aligned}
 \text{Predicted DE} &= (0 + 2 + 100 + 100 + 0) / 5 \\
 &= 40.4
 \end{aligned}$$

$$\begin{aligned}
 \text{Predicted DE} &= (3 + 100 + 20 + 10) / 4 \\
 &= 33.25
 \end{aligned}$$

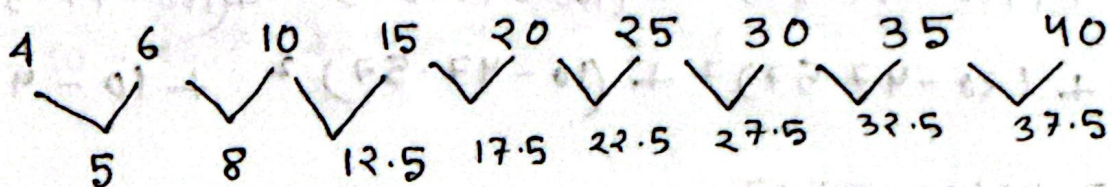
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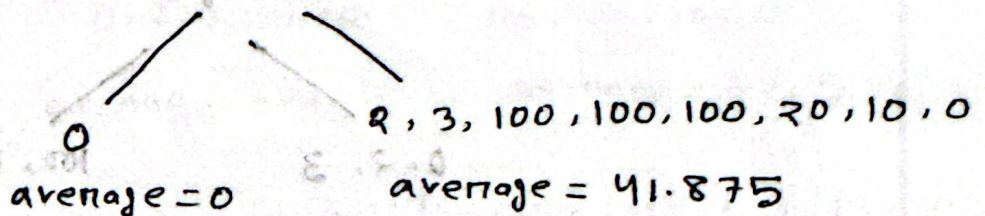
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$$\begin{aligned} SSR(\text{Gender}) &= (0 - 40.4)^2 + (2 - 40.4)^2 + (100 - 40.4)^2 \\ &+ (100 - 40.4)^2 + (0 - 40.4)^2 + (3 - 33.25)^2 + (100 - 33.25)^2 \\ &+ (20 - 33.25)^2 + (10 - 33.25)^2 \\ &= 17929.95 \end{aligned}$$

Dosage



Dosage < 5



$$\begin{aligned} SSR(\text{Dosage} < 5) &= (0 - 0)^2 + (2 - 41.875)^2 + (3 - 41.875)^2 \\ &+ (100 - 41.875)^2 + (100 - 41.875)^2 + (100 - 41.875)^2 \\ &+ (20 - 41.875)^2 + (10 - 41.875)^2 + (0 - 41.875)^2 \\ &= 16484.875 \end{aligned}$$

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Dosage < 8

0, 2

3, 100, 100, 100, 20, 10, 0

average = 1

average = 47.57

$$\begin{aligned} SSR(Dosage < 8) &= (0-1)^2 + (2-1)^2 + (3-47.57)^2 \\ &+ (100-47.57)^2 + (100-47.57)^2 + (100-47.57)^2 \\ &+ (20-47.57)^2 + (10-47.57)^2 + (0-47.57)^2 \\ &= 14669.7143 \end{aligned}$$

Dosage < 12.5

0, 2, 3

100, 100, 100, 20, 10, 0

average = 1.67

average = 55

$$\begin{aligned} SSR(Dosage < 12.5) &= (0-1.67)^2 + (2-1.67)^2 \\ &+ (3-1.67)^2 + (100-55)^2 + (100-55)^2 + (100-55)^2 \\ &+ (20-55)^2 + (10-55)^2 + (0-55)^2 \\ &= 12354.6667 \end{aligned}$$

GOODLUCK

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Dosage < 17.5

0, 100, 20, 3, 100

100, 100, 20, 10, 0

average = 26.25

= 46

$$SSR(\text{Dosage} < 17.5) = (0 - 26.25)^2 + (2 - 26.25)^2 + (3 - 26.25)^2 + (100 - 26.25)^2 + (100 - 46)^2 + (100 - 46)^2 + (20 - 46)^2 + (10 - 46)^2 + (0 - 46)^2 = 1717.72$$

Dosage < 22.5

0, 2, 3, 100, 100

100, 20, 10, 0

average = 41

average = 32.5

$$SSR(\text{Dosage} < 22.5) = (0 - 41)^2 + (2 - 41)^2 + (3 - 41)^2 + (100 - 41)^2 + (100 - 41)^2 + (100 - 32.5)^2 + (20 - 32.5)^2 + (10 - 32.5)^2 + (0 - 32.5)^2 = 17883$$

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Donage < 27.5

0, 2, 3, 100, 100, 100, 20, 10, 0

average = 50.83

average = 10

$$\begin{aligned} SSR(\text{Donage} < 27.5) &= (0 - 50.83)^2 + (2 - 50.83)^2 \\ &+ (3 - 50.83)^2 + (100 - 50.83)^2 + (100 - 50.83)^2 \\ &+ (100 - 50.83)^2 + (20 - 10)^2 + (10 - 10)^2 + (0 - 10)^2 \\ &= 14709 \end{aligned}$$

Donage < 32.5

0, 2, 3, 100, 100, 100, 20, 10, 0

average = 46.43

average = 5

$$\begin{aligned} SSR(\text{Donage} < 32.5) &= (0 - 46.43)^2 + (2 - 46.43)^2 \\ &+ (3 - 46.43)^2 + (100 - 46.43)^2 + (100 - 46.43)^2 \\ &+ (100 - 46.43)^2 + (20 - 46.43)^2 + (10 - 5)^2 + (0 - 5)^2 \\ &= 15374 \end{aligned}$$

GOOD LUCK

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Donage < 37.5

0, 2, 3, 100, 100,
100, 20, 10

0

average = 0

average = 41.875

$$\begin{aligned} SSR(\text{Donage} < 37.5) &= (0 - 41.875)^2 + (2 - 41.875)^2 \\ &+ (3 - 41.875)^2 + (100 - 41.875)^2 + (100 - 41.875)^2 \\ &+ (100 - 41.875)^2 + (20 - 41.875)^2 + (10 - 41.875)^2 \\ &+ (0 - 0)^2 \\ &= 16485 \end{aligned}$$

Threshold	SSR
5	16485
8	14670
12.5	12355 ✓
17.5	17177
22.5	17883
27.5	14709
32.5	15374
37.5	16485

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Dosage < 12.5

0, 2, 3

100, 100, 100, 20, 10, 0

Dosage	Gender	Drug effectiveness
17.5 < 15	F	100
20	M	100
22.5 < 20	M	100
25	F	20
27.5 < 25	F	10
30	F	10
32.5 < 30	F	0
35	M	0
37.5 < 35	M	0
40	M	0

Gender

M

F

100, 100, 0

100, 20, 10

Average = 66.67

average = 43.33

$$\begin{aligned}
 SSR(\text{Gender}) &= (100 - 66.67)^2 + (100 - 66.67)^2 \\
 &+ (0 - 66.67)^2 + (100 - 43.33)^2 + (20 - 43.33)^2 \\
 &+ (10 - 43.33)^2 \\
 &= 11533.33
 \end{aligned}$$

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Donage < 17.5

100

average = 100

100, 100, 20, 10, 0

average = 46

$$\begin{aligned} SSR(\text{donage} < 17.5) &= (100 - 100)^2 + (100 - 46)^2 \\ &+ (100 - 46)^2 + (20 - 46)^2 + (10 - 46)^2 + (0 - 46)^2 \\ &= 9920 \end{aligned}$$

Donage < 22.5

100, 100

average = 100

100, 20, 10, 0

average = 32.5

$$\begin{aligned} SSR(\text{donage} < 22.5) &= (100 - 100)^2 + (100 - 100)^2 \\ &+ (100 - 32.5)^2 + (20 - 32.5)^2 + (10 - 32.5)^2 \\ &+ (0 - 32.5)^2 \\ &= 6275 \end{aligned}$$

Donage < 27.5

100, 100, 100

average = 100

20, 10, 0

average = 10

$$\begin{aligned} SSR(\text{donage} < 27.5) &= (100 - 100)^2 + (100 - 100)^2 + (100 - 100)^2 \\ &+ (20 - 10)^2 + (10 - 10)^2 + (0 - 10)^2 = 200 \end{aligned}$$

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2. F1 > Donage < 32.5

100, 100, 100, 20

average = 80

10, 0

average = 5

$$\begin{aligned} SSR(\text{donage} < 32.5) &= (100 - 80)^2 + (100 - 80)^2 \\ &+ (100 - 80)^2 + (20 - 80)^2 + (10 - 5)^2 + (0 - 5)^2 \\ &= 4850 \end{aligned}$$

Donage < 37.5

100, 100, 100, 20, 10

average = 66

$$\begin{aligned} SSR(\text{donage} < 37.5) &= (100 - 66)^2 + (100 - 66)^2 \\ &+ (100 - 66)^2 + (20 - 66)^2 + (10 - 66)^2 + (0 - 66)^2 \\ &= 8720 \end{aligned}$$

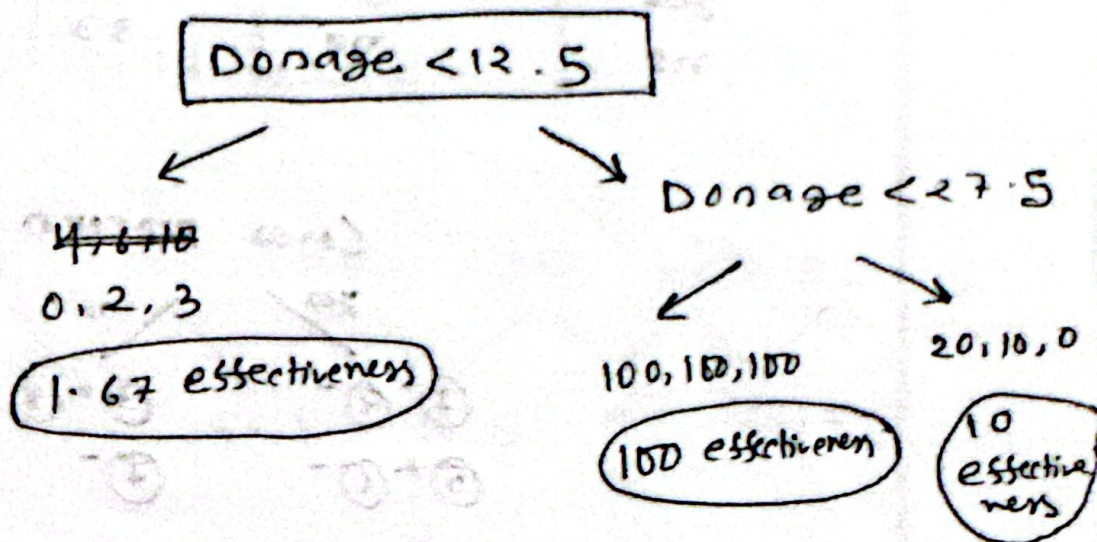
8720

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Threshold	SSR
17.5	9920
22.5	6275
27.5	200 ✓
32.5	4850
37.5	8720



$$2FE = 2 \left(\frac{1}{2} \right) - 2 \left(\frac{1}{4} \right) - 1 = (0.5) \text{ and}$$

$$FE = 2 \left(\frac{1}{4} \right) - 2 \left(\frac{1}{8} \right) - 1 = (0.25) \text{ and}$$

$$FE = 2 \left(\frac{1}{8} \right) - 2 \left(\frac{1}{16} \right) - 1 = (0.125) \text{ and}$$

$$2FE = 2 \left(\frac{1}{16} \right) - 2 \left(\frac{1}{32} \right) - 1 = (0.0625) \text{ and}$$